Question ID: 83f2c3bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$$y = x + 4$$

Which table gives three values of x and their corresponding values of y for the given equation?

Answer

A.

x	y
0	4
1	5
2	6

B.

x	y
0	6
1	5
2	4

C.

x	y
0	2
1	1
2	0

D.

x	y
0	0
1	1
2	2

Correct Answer: A

Rationale

Choice A is correct. Substituting 0 for x into the given equation yields $y = 0 + 4$, or $y = 4$. Therefore, when $x = 0$, the corresponding value of y for the given equation is 4. Substituting 1 for x into the given equation yields $y = 1 + 4$, or $y = 5$. Therefore, when $x = 1$, the corresponding value of y for the given equation is 5. Substituting 2 for x into the given equation yields $y = 2 + 4$, or $y = 6$. Therefore, when $x = 2$, the corresponding value of y for the given equation is 6. Of the choices given, only the table in choice A gives these three values of x and their corresponding values of y for the given equation.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 6$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 2$.

Choice D is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = x$.

Question ID: 4d8ccb96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A chemist studying the impact of salt on a process mixes x kilograms of a low-salt mixture, which is 2% salt by weight, with y kilograms of a high-salt mixture, which is 96% salt by weight, to create 24 kilograms of a mixture that is 4% salt by weight. Which equation represents this situation?

Answer

A. $0.96x + 0.02y = (0.04)(24)$

B. $0.02x + 0.96y = (0.04)(24)$

C. $0.96x + 0.02y = 24$

D. $0.02x + 0.96y = 24$

Correct Answer: B

Rationale

Choice B is correct. It's given that a chemist mixes x kilograms of a low-salt mixture, which is 2% salt by weight. Multiplying 0.02 by the amount of the low-salt mixture, x kilograms, yields $0.02x$ kilograms of salt in the low-salt mixture. It's also given that the chemist mixes y kilograms of a high-salt mixture, which is 96% salt by weight. Multiplying 0.96 by the amount of the high-salt mixture, y kilograms, yields $0.96y$ kilograms of salt in the high-salt mixture. Therefore, the total amount of salt in the combined mixture is $0.02x + 0.96y$ kilograms. It's given that the low-salt mixture and the high-salt mixture together create 24 kilograms of a combined mixture that is 4% salt by weight. Thus, the amount of salt in the combined mixture is $0.04(24)$ kilograms. Since the total amount of salt in the combined mixture equals the amount of salt in the low-salt mixture and the amount of salt in the high-salt mixture, the equation $0.02x + 0.96y = (0.04)(24)$ represents this situation.

Choice A is incorrect. This equation represents a situation where the low-salt mixture is 96%, not 2%, salt by weight and the high-salt mixture is 2%, not 96%, salt by weight.

Choice C is incorrect. This equation represents a situation where the low-salt mixture is 96%, not 2%, salt by weight and the high-salt mixture is 2%, not 96%, salt by weight, and where the combined mixture contains 24 kilograms of salt, not 24 kilograms of a mixture that is 4% salt by weight.

Choice D is incorrect. This equation represents a situation where the combined mixture contains 24 kilograms of salt, not 24 kilograms of a mixture that is 4% salt by weight.

Question ID: b23bba4c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$3a+4b=25$

A shipping company charged a customer $25 to ship some small boxes and some large boxes. The equation above represents the relationship between a, the number of small boxes, and b, the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

Answer

A. 3

B. 4

C. 5

D. 6

Correct Answer: B

Rationale

Choice B is correct. It's given that a represents the number of small boxes and b represents the number of large boxes the customer had shipped. If the customer had 3 small boxes shipped, then $a=3$. Substituting 3 for a in the equation $3a+4b=25$ yields $3(3)+4b=25$ or $9+4b=25$. Subtracting 9 from both sides of the equation yields $4b=16$. Dividing both sides of this equation by 4 yields $b=4$. Therefore, the customer had 4 large boxes shipped.

Choices A, C, and D are incorrect. If the number of large boxes shipped is 3, then $b=3$. Substituting 3 for b in the given equation yields $3a+4(3)=25$ or $3a+12=25$. Subtracting 12 from both sides of the equation and then dividing by 3 yields $a=\frac{13}{3}$. However, it's given that the number of small boxes shipped, a, is 3, not $\frac{13}{3}$, so b cannot equal 3. Similarly, if $b=5$ or $b=6$, then $a=\frac{5}{3}$ or $a=\frac{1}{3}$, respectively, which is also not true.

Question ID: 87322577

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$x + y = 75$

The equation above relates the number of minutes, x, Maria spends running each day and the number of minutes, y, she spends biking each day. In the equation, what does the number 75 represent?

Answer

A. The number of minutes spent running each day

B. The number of minutes spent biking each day

C. The total number of minutes spent running and biking each day

D. The number of minutes spent biking for each minute spent running

Correct Answer: C

Rationale

Choice C is correct. Maria spends x minutes running each day and y minutes biking each day. Therefore, $x + y$ represents the total number of minutes Maria spent running and biking each day. Because $x + y = 75$, it follows that 75 is the total number of minutes that Maria spent running and biking each day.

Choices A and B are incorrect. The number of minutes Maria spent running each day is represented by x and need not be 75. Similarly, the number of minutes that Maria spends biking each day is represented by y and need not be 75. The number of minutes Maria spends running each day and biking each day may vary; however, the total number of minutes she spends each day on these activities is constant and equal to 75. Choice D is incorrect. The number of minutes Maria spent biking for each minute spent running cannot be determined from the information provided.

Question ID: 6fa1dc0f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

Line r in the *xy*-plane has a slope of 4 and passes through the point $(0, 6)$. Which equation defines line r?

Answer

A. $y = -6x + 4$

B. $y = 6x + 4$

C. $y = 4x - 6$

D. $y = 4x + 6$

Correct Answer: D

Rationale

Choice D is correct. A line in the *xy*-plane with a slope of m and a *y*-intercept of $(0, b)$ can be defined by an equation in the form $y = mx + b$. It's given that line r has a slope of 4 and passes through the point $(0, 6)$. It follows that $m = 4$ and $b = 6$. Substituting 4 for m and 6 for b in the equation $y = mx + b$ yields $y = 4x + 6$. Therefore, the equation $y = 4x + 6$ defines line r.

Choice A is incorrect. This equation defines a line that has a slope of -6, not 4, and passes through the point $(0, 4)$, not $(0, 6)$.

Choice B is incorrect. This equation defines a line that has a slope of 6, not 4, and passes through the point $(0, 4)$, not $(0, 6)$.

Choice C is incorrect. This equation defines a line that passes through the point $(0, -6)$, not $(0, 6)$.

Question ID: c6b151d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A total of 364 paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. What is the best interpretation of $(x, y) = (24, 73)$ in this context?

Answer

A. If 24 triangles were constructed, then 73 rectangles were constructed.

B. If 24 triangles were constructed, then 73 paper straws were used.

C. If 73 triangles were constructed, then 24 rectangles were constructed.

D. If 73 triangles were constructed, then 24 paper straws were used.

Correct Answer: A

Rationale

Choice A is correct. It's given that 364 paper straws of equal length were used to construct triangles and rectangles, where no two polygons had a common side. It's also given that the equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. The equation $(x, y) = (24, 73)$ means that if $x = 24$, then $y = 73$. Substituting 24 for x and 73 for y in $3x + 4y = 364$ yields $3(24) + 4(73) = 364$, or $364 = 364$, which is true. Therefore, in this context, the equation $(x, y) = (24, 73)$ means that if 24 triangles were constructed, then 73 rectangles were constructed.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 8c98c834

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

The equation $y = 0.1x$ models the relationship between the number of different pieces of music a certain pianist practices, y, during an x-minute practice session. How many pieces did the pianist practice if the session lasted 30 minutes?

Answer

A. 1

B. 3

C. 10

D. 30

Correct Answer: B

Rationale

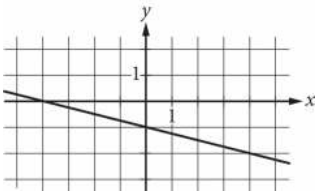
Choice B is correct. It's given that the equation $y = 0.1x$ models the relationship between the number of different pieces of music a certain pianist practices, y, and the number of minutes in a practice session, x. Since it's given that the session lasted 30 minutes, the number of pieces the pianist practiced can be found by substituting 30 for x in the given equation, which yields $y = 0.1(30)$, or $y = 3$.

Choices A and C are incorrect and may result from misinterpreting the values in the equation. Choice D is incorrect. This is the given value of x, not the value of y.

Question ID: b2845d88

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



Which of the following is an equation of the graph shown in the xy-plane above?

Answer

A. $y = -\frac{1}{4}x - 1$

B. $y = -x - 4$

C. $y = -x - \frac{1}{4}$

D. $y = -4x - 1$

Correct Answer: A

Rationale

Choice A is correct. The slope of the line can be found by choosing any two points on the line, such as (4, –2) and (0, –1). Subtracting the y-values results in –2 – (–1) = –1, the change in y. Subtracting the x-values results in 4 – 0 = 4, the change in *x*. Dividing the change in y by the change in *x* yields $-1 \div 4 = -\frac{1}{4}$, the slope. The line intersects the y-axis at (0, –1), so –1 is the y-coordinate of the y-intercept. This information can be expressed in slope-intercept form as the equation $y = -\frac{1}{4}x - 1$.

Choice B is incorrect and may result from incorrectly calculating the slope and then misidentifying the slope as the y-intercept. Choice C is incorrect and may result from misidentifying the slope as the y-intercept. Choice D is incorrect and may result from incorrectly calculating the slope.

Question ID: 8adf1335

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A city's total expense budget for one year was x million dollars. The city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between x and y in this context?

Answer

A. $x+y=201$

B. $x-y=201$

C. $2x-y=201$

D. $y-x=201$

Correct Answer: B

Rationale

Choice B is correct. Of the city's total expense budget for one year, the city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. This means that the expression $y+201$ represents the total expense budget, in millions of dollars, for one year. It's given that the total expense budget for one year is x million dollars. It follows then that the expression $y+201$ is equivalent to x, or $y+201=x$. Subtracting y from both sides of this equation yields $201=x-y$. By the symmetric property of equality, this is the same as $x-y=201$.

Choices A and C are incorrect. Because it's given that the total expense budget for one year, x million dollars, is comprised of the departmental expenses, y million dollars, and all other expenses, 201 million dollars, the expressions $x+y$ and $2x-y$ both must be equivalent to a value greater than 201 million dollars. Therefore, the equations $x+y=201$ and $2x-y=201$ aren't true. Choice D is incorrect. The value of x must be greater than the value of y. Therefore, $y-x=201$ can't represent this relationship.

Question ID: e7343559

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$$y = -4x + 40$$

Which table gives three values of x and their corresponding values of y for the given equation?

Answer

A.

x	y
0	0
1	-4
2	-8

B.

x	y
0	40
1	44
2	48

C.

x	y
0	40
1	36
2	32

D.

x	y
0	0
1	4
2	8

Correct Answer: C

Rationale

Choice C is correct. Each of the given choices gives three values of x: 0, 1, and 2. Substituting 0 for x in the given equation yields $y = -4(0) + 40$, or $y = 40$. Therefore, when $x = 0$, the corresponding value of y for the given equation is 40. Substituting 1 for x in the given equation yields $y = -4(1) + 40$, or $y = 36$. Therefore, when $x = 1$, the corresponding value of y for the given equation is 36. Substituting 2 for x in the given equation yields $y = -4(2) + 40$, or $y = 32$. Therefore, when $x = 2$, the corresponding value of y for the given equation is 32. Choice C gives three values of x, 0, 1, and 2, and their corresponding values of y, 40, 36, and 32, respectively, for the given equation.

Choice A is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -4x$.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x + 40$.

Choice D is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x$.

Question ID: dd797fe2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$4x + 3y = 24$

Mario purchased 4 binders that cost x dollars each and 3 notebooks that cost y dollars each. If the given equation represents this situation, which of the following is the best interpretation of 24 in this context?

Answer

A. The total cost, in dollars, for all binders purchased

B. The total cost, in dollars, for all notebooks purchased

C. The total cost, in dollars, for all binders and notebooks purchased

D. The difference in the total cost, in dollars, between the number of binders and notebooks purchased

Correct Answer: C

Rationale

Choice C is correct. Since Mario purchased 4 binders that cost x dollars each, the expression $4x$ represents the total cost, in dollars, of the 4 binders he purchased. Since Mario purchased 3 notebooks that cost y dollars each, the expression $3y$ represents the total cost, in dollars, of the 3 notebooks he purchased. Therefore, the expression $4x + 3y$ represents the total cost, in dollars, for all binders and notebooks he purchased. In the given equation, the expression $4x + 3y$ is equal to 24. Therefore, it follows that 24 is the total cost, in dollars, for all binders and notebooks purchased.

Choice A is incorrect. This is represented by the expression $4x$ in the given equation. Choice B is incorrect. This is represented by the expression $3y$ in the given equation. Choice D is incorrect. This is represented by the expression $|4x - 3y|$.

Question ID: 789975b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A gardener buys two kinds of fertilizer. Fertilizer A contains 60% filler materials by weight and Fertilizer B contains 40% filler materials by weight. Together, the fertilizers bought by the gardener contain a total of 240 pounds of filler materials. Which equation models this relationship, where x is the number of pounds of Fertilizer A and y is the number of pounds of Fertilizer B?

Answer

A. $0.4x + 0.6y = 240$

B. $0.6x + 0.4y = 240$

C. $40x + 60y = 240$

D. $60x + 40y = 240$

Correct Answer: B

Rationale

Choice B is correct. Since Fertilizer A contains 60% filler materials by weight, it follows that x pounds of Fertilizer A consists of 0.6x pounds of filler materials. Similarly, y pounds of Fertilizer B consists of 0.4y pounds of filler materials. When x pounds of Fertilizer A and y pounds of Fertilizer B are combined, the result is 240 pounds of filler materials. Therefore, the total amount, in pounds, of filler materials in a mixture of x pounds of Fertilizer A and y pounds of Fertilizer B can be expressed as $0.6x + 0.4y = 240$.

Choice A is incorrect. This choice transposes the percentages of filler materials for Fertilizer A and Fertilizer B. Fertilizer A consists of 0.6x pounds of filler materials and Fertilizer B consists of 0.4y pounds of filler materials. Therefore, $0.6x + 0.4y$ is equal to 240, not $0.4x + 0.6y$. Choice C is incorrect. This choice transposes the percentages of filler materials for Fertilizer A and Fertilizer B and incorrectly represents how to take the percentage of a value mathematically. Choice D is incorrect. This choice incorrectly represents how to take the percentage of a value mathematically. Fertilizer A consists of 0.6x pounds of filler materials, not 60x pounds of filler materials, and Fertilizer B consists of 0.4y pounds of filler materials, not 40y pounds of filler materials.

Question ID: cea27ab2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$$7x - 4y = -84$$

For the given equation, which table gives three values of x and their corresponding values of y?

Answer

A.

x	0	4	8
y	21	28	35

B.

x	0	4	8
y	35	28	21

C.

x	21	28	35
y	0	4	8

D.

x	21	28	35
y	8	4	0

Correct Answer: A

Rationale

Choice A is correct. To verify which table represents this linear relationship, the values in each table can be checked by substituting them into the given equation. The table in choice A shows that when $x = 0$, $y = 21$. Substituting these values into the given equation yields $7(0) - 4(21) = -84$, or $-84 = -84$, which is true. Additionally, the table in choice A shows that when $x = 4$, $y = 28$. Substituting these values into the given equation yields $7(4) - 4(28) = -84$, or $-84 = -84$, which is true. Finally, the table in choice A shows that when $x = 8$, $y = 35$. Substituting these values into the given equation yields $7(8) - 4(35) = -84$, or $-84 = -84$, which is true. Therefore, the table in choice A gives three values of x and their corresponding values of y.

Choice B is incorrect. The table in choice B shows that when $x = 0$, $y = 35$. Substituting these values into the given equation yields $7(0) - 4(35) = -84$, or $-140 = -84$, which is not true.

Choice C is incorrect. The table in choice C shows that when $x = 21$, $y = 0$. Substituting these values into the given equation yields $7(21) - 4(0) = -84$, or $147 = -84$, which is not true.

Choice D is incorrect. The table in choice D shows that when $x = 21$, $y = 8$. Substituting these values into the given equation yields $7(21) - 4(8) = -84$, or $115 = -84$, which is not true.

Question ID: 2554b413

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

In the xy-plane, a line has a slope of 6 and passes through the point $(0,8)$. Which of the following is an equation of this line?

Answer

A. $y=6x+8$

B. $y=6x+48$

C. $y=8x+6$

D. $y=8x+48$

Correct Answer: A

Rationale

Choice A is correct. The slope-intercept form of an equation for a line is $y=mx+b$, where m is the slope of the line and b is the y-coordinate of the y-intercept of the line. It's given that the slope is 6, so $m=6$. It's also given that the line passes through the point $(0,8)$ on the y-axis, so $b=8$. Substituting $m=6$ and $b=8$ into the equation $y=mx+b$ gives $y=6x+8$.

Choices B, C, and D are incorrect and may result from misinterpreting the slope-intercept form of an equation of a line.

Question ID: ba79f10f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

x	y
0	18
1	13
2	8

The table shows three values of x and their corresponding values of y. There is a linear relationship between x and y. Which of the following equations represents this relationship?

Answer

A. $y = 18x + 13$

B. $y = 18x + 18$

C. $y = -5x + 13$

D. $y = -5x + 18$

Correct Answer: D

Rationale

Choice D is correct. A linear relationship can be represented by an equation of the form $y = mx + b$, where m and b are constants. It's given in the table that when $x = 0$, $y = 18$. Substituting 0 for x and 18 for y in $y = mx + b$ yields $18 = m(0) + b$, or $18 = b$. Substituting 18 for b in the equation $y = mx + b$ yields $y = mx + 18$. It's also given in the table that when $x = 1$, $y = 13$. Substituting 1 for x and 13 for y in the equation $y = mx + 18$ yields $13 = m(1) + 18$, or $13 = m + 18$. Subtracting 18 from both sides of this equation yields $-5 = m$. Therefore, the equation $y = -5x + 18$ represents the relationship between x and y.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: dfa45424

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

Tony spends $80 per month on public transportation. A 10-ride pass costs $12.50, and a single-ride pass costs $1.50. If g represents the number of 10-ride passes Tony buys in a month and t represents the number of single-ride passes Tony buys in a month, which of the following equations best represents the relationship between g and t ?

Answer

A. $g+t=80$

B. $g+t=1.50+12.50$

C. $1.50g+12.50t=80$

D. $12.50g+1.50t=80$

Correct Answer: D

Rationale

Choice D is correct. Since a 10-ride pass costs $12.50 and g is the number of 10-ride passes Tony buys in a month, the expression $12.50g$ represents the amount Tony spends on 10-ride passes in a month. Since a single-ride pass costs $1.50 and t is the number of single-ride passes Tony buys in a month, the expression $1.50t$ represents the amount Tony spends on single-ride passes in a month. Therefore, the sum $12.50g+1.50t$ represents the amount he spends on the two types of passes in a month. Since Tony spends a total of $80 on passes in a month, this expression can be set equal to 80, producing $12.50g+1.50t=80$.

Choices A and B are incorrect. The expression $g+t$ represents the total number of the two types of passes Tony buys in a month, not the amount Tony spends, which is equal to 80, nor the cost of one of each pass, which is equal to $1.50+12.50$. Choice C is incorrect and may result from reversing the cost for each type of pass Tony buys in a month.

Question ID: 10c448d6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A line in the *xy*-plane has a slope of $\frac{1}{9}$ and passes through the point $(0, 14)$. Which equation represents this line?

Answer

A. $y = -\frac{1}{9}x - 14$

B. $y = -\frac{1}{9}x + 14$

C. $y = \frac{1}{9}x - 14$

D. $y = \frac{1}{9}x + 14$

Correct Answer: D

Rationale

Choice D is correct. The equation of a line in the *xy*-plane can be written as $y = mx + b$, where m represents the slope of the line and $(0, b)$ represents the *y*-intercept of the line. It's given that the slope of the line is $\frac{1}{9}$. It follows that $m = \frac{1}{9}$. It's also given that the line passes through the point $(0, 14)$. It follows that $b = 14$. Substituting $\frac{1}{9}$ for m and 14 for b in $y = mx + b$ yields $y = \frac{1}{9}x + 14$. Thus, the equation $y = \frac{1}{9}x + 14$ represents this line.

Choice A is incorrect. This equation represents a line with a slope of $-\frac{1}{9}$ and a *y*-intercept of $(0, -14)$.

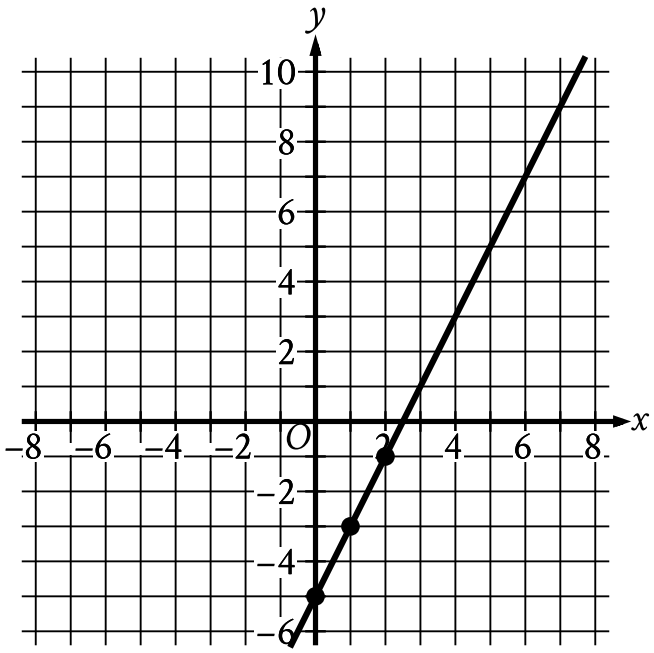
Choice B is incorrect. This equation represents a line with a slope of $-\frac{1}{9}$ and a *y*-intercept of $(0, 14)$.

Choice C is incorrect. This equation represents a line with a slope of $\frac{1}{9}$ and a *y*-intercept of $(0, -14)$.

Question ID: 4acd05cd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



The graph shows the linear relationship between x and y. Which table gives three values of x and their corresponding values of y for this relationship?

Answer

A.

x	y
0	0
1	−7
2	−9

B.

x	y
0	0
1	−3
2	−1

C.

x	y
0	−5
1	−7
2	−9

D.

x	y
0	−5
1	−3

2	-1

Correct Answer: D

Rationale

Choice D is correct. It's given that the graph shows the linear relationship between x and y. The given graph passes through the points $(0, -5)$, $(1, -3)$, and $(2, -1)$. It follows that when $x = 0$, the corresponding value of y is -5, when $x = 1$, the corresponding value of y is -3, and when $x = 2$, the corresponding value of y is -1. Of the given choices, only the table in choice D gives these three values of x and their corresponding values of y for the relationship shown in the graph.

Choice A is incorrect. This table represents a relationship between x and y such that the graph passes through the points $(0, 0)$, $(1, -7)$, and $(2, -9)$.

Choice B is incorrect. This table represents a relationship between x and y such that the graph passes through the points $(0, 0)$, $(1, -3)$, and $(2, -1)$.

Choice C is incorrect. This table represents a linear relationship between x and y such that the graph passes through the points $(0, -5)$, $(1, -7)$, and $(2, -9)$.

Question ID: b2de69bd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

x	y
1	5
2	7
3	9
4	11

The table above shows some pairs of x values and y values. Which of the following equations could represent the relationship between x and y ?

Answer

A. $y = 2x + 3$

B. $y = 3x - 2$

C. $y = 4x - 1$

D. $y = 5x$

Correct Answer: A

Rationale

Choice A is correct. Each of the choices is a linear equation in the form y = mx + b, where m and b are constants. In this equation, m represents the change in y for each increase in x by 1. From the table, it can be determined that the value of y increases by 2 for each increase in x by 1. In other words, for the pairs of x and y in the given table, m = 2. The value of b can be found by substituting the values of x and y from any row of the table and substituting the value of m into the equation y = mx + b and then solving for b. For example, using x = 1, y = 5, and m = 2 yields 5 = 2(1) + b. Solving for b yields b = 3. Therefore, the equation y = 2x + 3 could represent the relationship between x and y in the given table.

Alternatively, if an equation represents the relationship between x and y, then when each pair of x and y from the table is substituted into the equation, the result will be a true statement. Of the equations given, the equation y = 2x + 3 in choice A is the only equation that results in a true statement when each of the pairs of x and y are substituted into the equation.

Choices B, C, and D are incorrect because when at least one pair of x and y from the table is substituted into the equations given in these choices, the result is a false statement. For example, when the pair x = 4 and y = 11 is substituted into the equation in choice B, the result is 11 = 3(4) – 2, or 11 = 10, which is false.

Question ID: 029c2dc2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A teacher is creating an assignment worth 70 points. The assignment will consist of questions worth 1 point and questions worth 3 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 3-point questions?

Answer

A. $4xy = 70$

B. $4(x + y) = 70$

C. $3x + y = 70$

D. $x + 3y = 70$

Correct Answer: D

Rationale

Choice D is correct. Since x represents the number of 1-point questions and y represents the number of 3-point questions, the assignment is worth a total of $1 \cdot x + 3 \cdot y$, or $x + 3y$, points. Since the assignment is worth 70 points, the equation $x + 3y = 70$ represents this situation.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: 1efd8202

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$$y = 70x + 8$$

Which table gives three values of x and their corresponding values of y for the given equation?

Answer

A.

x	y
0	8
2	148
4	288

B.

x	y
0	70
2	78
4	86

C.

x	y
0	70
2	140
4	280

D.

x	y
0	8
2	132
4	272

Correct Answer: A

Rationale

Choice A is correct. Each of the given choices gives three values of x: 0, 2, and 4. Substituting 0 for x in the given equation yields $y = 70(0) + 8$, or $y = 8$. Therefore, when $x = 0$, the corresponding value of y for the given equation is 8. Substituting 2 for x in the given equation yields $y = 70(2) + 8$, or $y = 148$. Therefore, when $x = 2$, the corresponding value of y for the given equation is 148. Substituting 4 for x in the given equation yields $y = 70(4) + 8$, or $y = 288$. Therefore, when $x = 4$, the corresponding value of y for the given equation is 288. Thus, if the three values of x are 0, 2, and 4, then their corresponding values of y are 8, 148, and 288, respectively, for the given equation.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x + 70$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: ed856e9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

What is the *y*-intercept of the graph of $y = 34x + 81$ in the *xy*-plane?

Answer

A. $(0, 81)$

B. $(0, 34)$

C. $(0, -34)$

D. $(0, -81)$

Correct Answer: A

Rationale

Choice A is correct. In the *xy*-plane, the graph of an equation in the form $y = mx + b$, where m and b are constants, has a slope of m and a *y*-intercept of $(0, b)$. Therefore, the *y*-intercept of the graph of $y = 34x + 81$ is $(0, 81)$.

Choice B is incorrect and may result from conceptual or calculation errors.

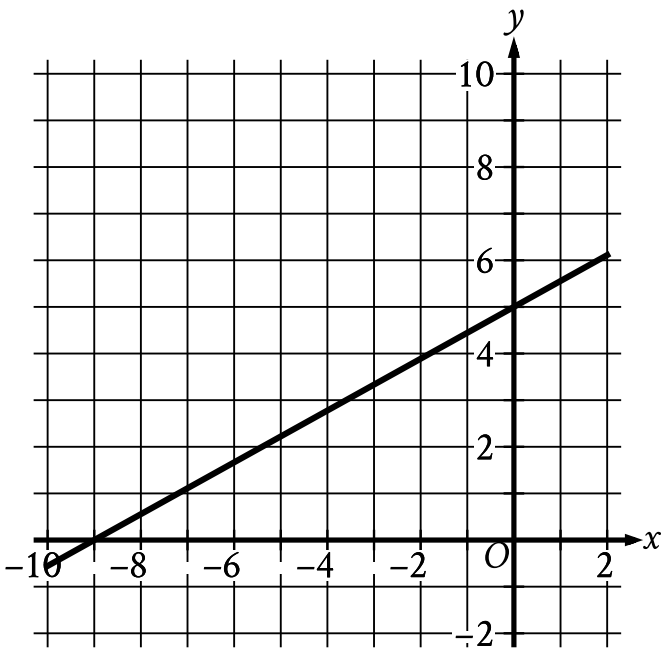
Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: eeebe166

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



What is the *y*-intercept of the line graphed?

Answer

A. $(-5, 0)$

B. $(0, 0)$

C. $(0, 5)$

D. $(0, 9)$

Correct Answer: C

Rationale

Choice C is correct. The *y*-intercept of a graph is the point where the graph intersects the *y*-axis. The line graphed intersects the *y*-axis at the point $(0, 5)$. Therefore, the *y*-intercept of the line graphed is $(0, 5)$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: b9839f9e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$F = 2.50x + 7.00y$

In the equation above, F represents the total amount of money, in dollars, a food truck charges for x drinks and y salads. The price, in dollars, of each drink is the same, and the price, in dollars, of each salad is the same. Which of the following is the best interpretation for the number 7.00 in this context?

Answer

A. The price, in dollars, of one drink

B. The price, in dollars, of one salad

C. The number of drinks bought during the day

D. The number of salads bought during the day

Correct Answer: B

Rationale

Choice B is correct. It's given that $2.50x + 7.00y$ is equal to the total amount of money, in dollars, a food truck charges for x drinks and y salads. Since each salad has the same price, it follows that the total charge for y salads is $7.00y$ dollars. When $y = 1$, the value of the expression $7.00y$ is 7.00×1, or 7.00. Therefore, the price for one salad is 7.00 dollars.

Choice A is incorrect. Since each drink has the same price, it follows that the total charge for x drinks is $2.50x$ dollars. Therefore, the price, in dollars, for one drink is 2.50, not 7.00. Choices C and D are incorrect. In the given equation, F represents the total charge, in dollars, when x drinks and y salads are bought at the food truck. No information is provided about the number of drinks or the number of salads that are bought during the day. Therefore, 7.00 doesn't represent either of these quantities.

Question ID: 1fe778dc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A line in the *xy*-plane has a slope of $-\frac{1}{2}$ and passes through the point $(0, 3)$. Which equation represents this line?

Answer

A. $y = -\frac{1}{2}x - 3$

B. $y = -\frac{1}{2}x + 3$

C. $y = \frac{1}{2}x - 3$

D. $y = \frac{1}{2}x + 3$

Correct Answer: B

Rationale

Choice B is correct. A line in the *xy*-plane with a slope of m and a *y*-intercept of $(0, b)$ can be represented by the equation $y = mx + b$. It's given that the line has a slope of $-\frac{1}{2}$. Therefore, $m = -\frac{1}{2}$. It's also given that the line passes through the point $(0, 3)$. Therefore, $b = 3$. Substituting $-\frac{1}{2}$ for m and 3 for b in the equation $y = mx + b$ yields $y = -\frac{1}{2}x + 3$. Therefore, the equation $y = -\frac{1}{2}x + 3$ represents this line.

Choice A is incorrect. This equation represents a line in the *xy*-plane that passes through the point $(0, -3)$, not $(0, 3)$.

Choice C is incorrect. This equation represents a line in the *xy*-plane that has a slope of $\frac{1}{2}$, not $-\frac{1}{2}$, and passes through the point $(0, -3)$, not $(0, 3)$.

Choice D is incorrect. This equation represents a line in the *xy*-plane that has a slope of $\frac{1}{2}$, not $-\frac{1}{2}$.

Question ID: 39617468

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$$x + y = 350$$

The given equation relates the total number of maple trees, x, and the total number of birch trees, y, planted in a 14-acre forest preserve. If 245 maple trees were planted in the forest preserve, how many birch trees were planted in the forest preserve?

Answer

A. 14

B. 25

C. 105

D. 245

Correct Answer: C

Rationale

Choice C is correct. It's given that the equation $x + y = 350$ relates the total number of maple trees, x, and the total number of birch trees, y, planted in a 14-acre forest preserve. It's also given that 245 maple trees were planted in the forest preserve. Substituting 245 for x in the given equation yields $245 + y = 350$. Subtracting 245 from both sides of this equation yields $y = 105$. Therefore, 105 birch trees were planted in the forest preserve.

Choice A is incorrect. This is the number of acres in the forest preserve, not the number of birch trees planted in the forest preserve.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the number of maple trees planted in the forest preserve, not the number of birch trees planted in the forest preserve.

Question ID: 8da536c6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

In 2010, a swim club had a total of 35 swimmers, each classified as either advanced or intermediate. From 2010 to 2020, the number of advanced swimmers in the club increased by approximately 53%, and the number of intermediate swimmers in the club increased by approximately 44%. The total number of swimmers in the club increased by approximately 49%. Which equation best represents this situation, where a represents the number of advanced swimmers in the club in 2010 and b represents the number of intermediate swimmers in the club in 2010?

Answer

A. $1.53a + 1.49b = 35(1.44)$

B. $1.49a + 0.53b = 35(1.44)$

C. $1.53a + 1.44b = 35(1.49)$

D. $1.44a + 1.53b = 35(1.49)$

Correct Answer: C

Rationale

Choice C is correct. It's given that in 2010, a swim club had a total of 35 swimmers, each classified as either advanced or intermediate, and that a represents the number of advanced swimmers in 2010 and b represents the number of intermediate swimmers in 2010. It's also given that from 2010 to 2020, the number of advanced swimmers in the club increased by approximately 53% and the number of intermediate swimmers in the club increased by approximately 44%. Thus, in 2020, the approximate number of advanced swimmers in the club can be represented as $1.53a$ and the approximate number of intermediate swimmers in the club can be represented as $1.44b$. It's given that the total number of swimmers in the club increased by approximately 49% from 2010 to 2020. Since the club had 35 swimmers in 2010, it follows that the total number of swimmers in 2020 can be represented as $35(1.49)$. Since the sum of the number of advanced swimmers in 2020 and the number of intermediate swimmers in 2020 equals the total number of swimmers in 2020, the equation $1.53a + 1.44b = 35(1.49)$ best represents this situation.

Choice A is incorrect. This equation represents a situation where the number of intermediate swimmers in the club in 2020 increased by approximately 49%, not 44%, and the total number of swimmers in the club in 2020 increased by approximately 44%, not 49%.

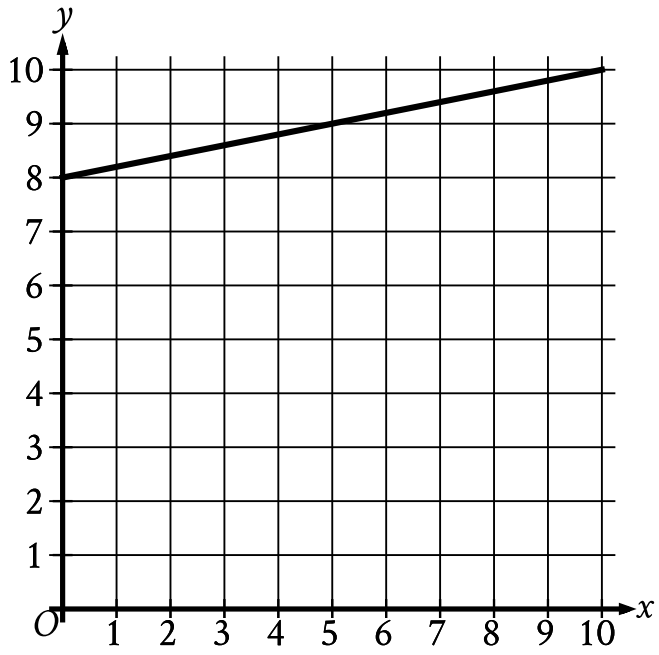
Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This equation represents a situation where the number of advanced swimmers in the club in 2020 increased by approximately 44%, not 53%, and the number of intermediate swimmers in the club in 2020 increased by approximately 53%, not 44%.

Question ID: f40552a9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



What is the *y*-intercept of the line graphed?

Answer

A. $(0, -8)$

B. $\left(0, -\frac{1}{8}\right)$

C. $(0, 0)$

D. $(0, 8)$

Correct Answer: D

Rationale

Choice D is correct. The *y*-intercept of a line graphed in the *xy*-plane is the point where the line intersects the *y*-axis. The line graphed intersects the *y*-axis at the point $(0, 8)$. Therefore, the *y*-intercept of the line graphed is $(0, 8)$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: 8b2a2a63

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

The *y*-intercept of the graph of $y = -6x - 32$ in the *xy*-plane is $(0, y)$. What is the value of y?

Correct Answer: -32

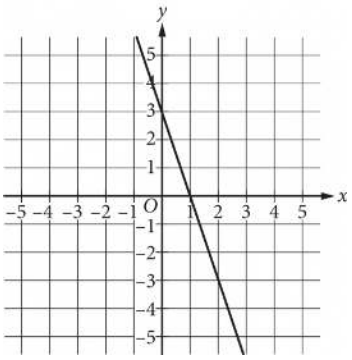
Rationale

The correct answer is -32. It's given that the *y*-intercept of the graph of $y = -6x - 32$ is $(0, y)$. Substituting 0 for x in this equation yields $y = -6(0) - 32$, or $y = -32$. Therefore, the value of y that corresponds to the *y*-intercept of the graph of $y = -6x - 32$ in the *xy*-plane is -32.

Question ID: 8a1544f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



What is the equation of the line shown in the xy-plane above?

Answer

A. $y = 3x - 3$

B. $y = -3x + 3$

C. $y = \frac{1}{3}x - 3$

D. $y = -\frac{1}{3}x + 3$

Correct Answer: B

Rationale

Choice B is correct. Any line in the xy-plane can be defined by an equation in the form y = mx + b, where m is the slope of the line and b is the y-coordinate of the y-intercept of the line. From the graph, the y-intercept of the line is (0, 3). Therefore, b = 3. The slope of the line is the change in the value of y divided by the change in the value of x for any two points on the line. The line shown passes through (0, 3) and (1, 0), so $m = \frac{3 - 0}{0 - 1}$, or m = −3. Therefore, the equation of the line is y = −3x + 3.

Choices A and C are incorrect because the equations given in these choices represent a line with a positive slope. However, the line shown has a negative slope. Choice D is incorrect because the equation given in this choice represents a line with slope of $-\frac{1}{3}$. However, the line shown has a slope of −3.